

ANCOMBC 2.15.1: computational changes and verification

Package maintainer notes

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1 Scope

Version 2.15.1 reduces the running time and the peak memory of the five user-facing entry points `ancom`, `ancombc`, `ancombc2`, `secom_linear` and `secom_dist`. It adds the argument `conservative` to `ancombc2`, which selects the rule used in the sensitivity analysis to pseudo-count addition. It corrects two defects: the failure of `ancombc` with `pseudo = 0` on a table containing zero counts, and the failure of `ancombc2` with `pseudo_sens = TRUE`, `trend = TRUE` and `global = FALSE`.

Three constraints apply to the computational changes. The implementation is R only, with no compiled code. No package is added to `Imports`, `Depends` or `Suggests`. The numerical results of version 2.13.2 are reproduced on every workload of the verification harness, at the tolerance stated in Section 2.

2 Verification procedure

The harness is `benchmarks/harness`, with five scripts: `workloads.R` defines the workloads, `run.R` times them and records peak memory, `compare.R` compares two sets of results, `edge_cases.R` exercises configurations that the workloads do not reach, and `repro_pseudo0.R` reproduces the `pseudo = 0` defect.

The harness contains 17 workloads. Twelve are the calls of the vignettes `ANCOM.Rmd`, `ANCOMBC.Rmd`, `ANCOMBC2.Rmd` and `SECOM.Rmd`, reproduced with their arguments, their seed `set.seed(123)` and their `n_cl` values. Five are larger configurations at the Genus or OTU level. Table 1 lists them. Sizes are the taxa and samples that enter the analysis after `prv_cut` and `lib_cut`.

All measurements were taken on one machine, an Apple M2 Max running R 4.5.3, with the two installations exercised back to back. Vignette-scale times are medians of three runs; large-scale times are single runs.

Equality criterion. Let a and b denote the result objects of the two versions for one workload. Components that carry an evaluation environment are replaced by a sentinel before comparison. The workload is recorded as equal when

$$\text{all.equal}(a, b, \text{tolerance} = 10^{-8}) = \text{TRUE}, \quad (1)$$

that is, when every pair of corresponding numeric components satisfies $|\overline{a - b}| / |\overline{a}| \leq 10^{-8}$ and every pair of non-numeric components is identical. The comparison is repeated at tolerance 0, which reports bitwise equality.

Determinism. Repeated runs of 2.13.2 on the 12 vignette workloads are bitwise identical, including the workloads with `n_cl = 2`, whose parallel streams are generated by `doRNG`.

Table 1: The 17 harness workloads. The last column is the 2.13.2 elapsed time.

Workload	Data	Level	Taxa $\times$ samples	Arguments	2.13.2 (s)
<i>Vignette scale</i>					
ancom_atlas	atlas1006	Family	21×873	main_var = "bmi", adj_formula = "age + region", n_cl = 2	2.84
ancom_dietswap	dietswap	Family	19×222	main_var = "group", rand_formula = "(timepoint subject)", n_cl = 2	6.14
ancombc_atlas	atlas1006	Family	21×873	formula = "age + region + bmi", global = TRUE, n_cl = 1	1.89
ancombc2_qmp	QMP simulation, $n = 150$	OTU	91×143	fix_formula = "cont_cov + cat_cov", pairwise = TRUE, pseudo_sens = TRUE, n_cl = 2	7.13
ancombc2_atlas_perm	atlas1006, permuted bmi	Genus	118×873	fix_formula = "bmi", prv_cut = 0, remaining arguments at their defaults	10.63
ancombc2_atlas_family	atlas1006	Family	21×873	fix_formula = "age + region + bmi", global, pairwise, Dunnett and trend tests, n_cl = 2	26.84
ancombc2_atlas_interact	atlas1006	Family	21×873	fix_formula = "age * bmi + region", n_cl = 2	24.99
ancombc2_dietswap	dietswap	Family	19×222	rand_formula = "(timepoint subject)", global, pairwise, Dunnett and trend tests, n_cl = 2	231.93
secom_linear_atlas	atlas1006	Phylum	6×873	method = "pearson", thresh_len = 20, n_cv = 10, n_cl = 2	1.38
secom_dist_atlas	atlas1006	Phylum	6×873	R = 1000, n_cl = 2	7.49
secom_linear_2groups	atlas1006, regions CE and NE	Phylum	6×578 , 6×181	thresh_len = 20, n_cv = 10, n_cl = 2	1.42
secom_dist_2groups	atlas1006, regions CE and NE	Phylum	6×578 , 6×181	R = 1000, n_cl = 2	4.84
<i>Large scale</i>					
ancom_atlas_genus_large	atlas1006	Genus	114×873	main_var = "bmi", adj_formula = "age + region", n_cl = 2	10.70
ancombc_atlas_genus_large	atlas1006	Genus	114×873	formula = "age + region + bmi", global = TRUE, n_cl = 1	10.37
ancombc2_atlas_genus_large	atlas1006	Genus	114×873	global, pairwise, Dunnett and trend tests, B = 10 for the trend bootstrap, n_cl = 2	29.57
ancombc2_qmp_otu_large	QMP simulation, $n = 900$	OTU	91×824	global, pairwise and Dunnett tests, struc_zero = TRUE, n_cl = 2	9.04
secom_linear_atlas_genus_large	atlas1006	Genus	114×873	prv_cut = 0.10, thresh_len = 100, n_cv = 10, n_cl = 2	7.95

Edge cases. `edge_cases.R` covers nine configurations: a non-finite response, a taxon whose usable samples cover one level of the group factor, an all-zero taxon, a data-dependent model term, an unused level of the group factor with and without `neg_lb`, a missing group label, a two-level group, and a covariate with missing values. Thirteen further `ancom` configurations vary the main variable type, the adjustment formula, the random-effects formula and `n_cl`. All are bitwise identical between the two versions, with the single exception of `ancombc(pseudo = 0)`, which is the defect described in Section 6.

Tests. The package tests are the `testthat` files for `ancom`, `ancombc`, `ancombc2` and `secom`, which include a test of `conservative = FALSE`. They are run by R CMD `check` on the built tarball, together with the five vignettes.

Memory measurement. Peak memory is the increase in the `gc()` “max used” figure of the master process, summed over `Ncells` and `Vcells`, between `gc(reset = TRUE)` before the call and `gc()` after it. Workloads with `n_cl > 1` spawn PSOCK worker processes whose allocations `gc()` does not observe. For those workloads the reported figure is a lower bound on the memory footprint of the run.

3 Results

Table 2 reports the back-to-back comparison of the 17 workloads. All 17 are equal at tolerance 10^{-8} . Fourteen are bitwise identical.

Table 2: Elapsed time, peak master-process memory and equality for the 17 workloads.

Workload	2.13.2 (s)	2.15.1 (s)	Speedup	2.13.2 (Mb)	2.15.1 (Mb)	Equality
<code>ancom.atlas</code>	2.84	1.37	2.08	58.2	45.3	bitwise
<code>ancom.atlas_genus_large</code>	10.70	1.42	7.54	79.3	33.3	bitwise
<code>ancom.dietswap</code>	6.14	3.48	1.76	48.8	50.3	bitwise
<code>ancombc.atlas</code>	1.89	0.49	3.85	345.4	342.3	bitwise
<code>ancombc.atlas_genus_large</code>	10.37	0.76	13.68	363.9	341.5	bitwise
<code>ancombc2.atlas_family</code>	26.84	3.41	7.87	294.1	111.2	bitwise
<code>ancombc2.atlas_genus_large</code>	29.57	3.61	8.19	294.0	168.4	bitwise
<code>ancombc2.atlas_interact</code>	24.99	3.49	7.17	219.6	111.0	bitwise
<code>ancombc2.atlas_perm</code>	10.63	0.89	12.01	369.3	211.9	bitwise
<code>ancombc2.dietswap</code>	231.93	64.63	3.59	356.3	309.5	bitwise
<code>ancombc2.qmp</code>	7.13	1.85	3.85	342.0	133.0	bitwise
<code>ancombc2.qmp_otu_large</code>	9.04	2.13	4.24	362.7	152.0	bitwise
<code>secom.dist_2groups</code>	4.84	3.52	1.38	128.2	116.5	bitwise
<code>secom.dist.atlas</code>	7.49	6.06	1.24	79.2	69.6	bitwise
<code>secom.linear_2groups</code>	1.42	0.35	4.09	125.6	119.1	10^{-8}
<code>secom.linear.atlas</code>	1.38	0.30	4.67	77.9	73.7	10^{-8}
<code>secom.linear.atlas_genus_large</code>	7.95	0.44	17.90	118.6	91.2	10^{-8}

The three `secom.linear` workloads differ in one component, `cv_error`, the cross-validation loss evaluated on the threshold grid. The mean relative difference is 1.40×10^{-16} for `secom.linear.atlas`, 2.63×10^{-16} for `secom.linear_2groups` and 3.34×10^{-16} for `secom.linear.atlas_genus_large`; the maximum absolute difference over the three workloads lies between 1.1×10^{-16} and 8.9×10^{-15} . The difference arises from the order of summation in the squared Frobenius norm. The quantities selected from `cv_error`, `thresh_opt` and `alpha_opt`, and the matrices `corr_th` and `corr_fl` derived from them, are identical.

R CMD `check` on the built tarball returns **Status:** OK, with 13 passing tests, no failures, no warnings and no skips, and all five vignettes re-built; the sensitivity analysis table of the rendered

ANCOMBC2 vignette reproduces the counts quoted in its section 3.4.

Table 3 reports configurations outside the harness that isolate the scaling of individual entry points.

Table 3: Synthetic and auxiliary scaling measurements. Times are single runs; memory is the master-process peak.

Function	Configuration	2.13.2 (s)	2.15.1 (s)	2.13.2 (Mb)	2.15.1 (Mb)
<code>ancom</code>	300 taxa, 100 samples	93.6	0.16	164.0	62.7
<code>ancom</code>	600 taxa, 100 samples	340.0	0.98	179.6	93.9
<code>ancom</code>	1000 taxa, 100 samples	881*	1.75		155.7
<code>ancombc</code>	2000 taxa, 200 samples	42.1	1.39	235.4	187.9
<code>ancombc2</code>	2000 taxa, 200 samples	75.5	4.94	327.6	247.2
<code>secom_dist</code>	114 taxa, 873 samples, $R = 50$	209.4	142.2	97.0	111.4

The synthetic configurations use a negative binomial count table with a three-level group variable, one continuous covariate and `n_cl = 1`, and for `ancombc2` the global, pairwise and Dunnett tests with `pseudo_sens = TRUE`. The entry marked * extrapolates the 2.13.2 time from the 300-taxon and 600-taxon times under the fitted power law in the number of taxa, with exponent 1.86. The `secom_dist` configuration is atlas1006 at the Genus level with `n_cl = 2`.

4 Changes by function

`ancom`

With `rand_formula = NULL`, `ancom` builds one design for all additive log-ratio responses. The model frame, the model matrix, the QR factorization, the residual degrees of freedom and the inverse cross-product matrix are computed once by `.ancom_alr_design` and shared by every pair of taxa. For one reference taxon, the responses of all remaining taxa form the columns of a response matrix passed to a single `stats::lm.fit` call. In `.ancom_alr_stats`, the F statistic of a categorical main variable is formed from the sequential sums of squares in `fit$effects`, as in `anova.lm`, and the t statistic of a continuous main variable from the coefficient on the second row of the design and the corresponding diagonal entry of the inverse cross-product matrix, as in `summary.lm`. The per-pair `lm()` path is used when the main variable is not a term of the model, when the design is rank deficient, when there are no residual degrees of freedom, or when a response is not finite.

Only the pairs with $j > i$ are fitted; the entry (j, i) is filled by reflection, with the sign of the t statistic and of the effect estimate reversed. With `rand_formula` supplied, each pair is fitted by `lme4::lmer`. The `foreach` loop iterates over `n_cl` interleaved blocks of reference taxa and exports the design list and the log-transformed table; the p-value and effect-size matrices are preallocated.

`ancombc` and `ancombc2`, fixed effects

`.lm_fit_all` groups taxa by their pattern of usable samples, where a sample is usable for a taxon when its response is finite and its design row is complete. Each group is solved by one multi-response `stats::lm.fit` on the shared design restricted to the usable samples. A group whose restricted design is rank deficient is refitted taxon by taxon with `stats::lm()`, so that an absent factor level is dropped in the same way. The pattern keys are built by a vectorized paste over the columns of the usability indicator.

`.sandwich_vcov` stores the per-sample outer products $x_j x_j^\top$, which do not depend on the taxon, once as the rows of an $n \times p^2$ matrix. The sum over samples is accumulated for all taxa in a block at a time, in the sample order used for the per-taxon accumulation, so the floating-point result is

unchanged. The block size bounds the size of the temporaries.

`.bias_em` evaluates `dnorm` on the whole coefficient vector in the E step, and builds the `nloptr` option list once per EM run.

The preparation functions compute prevalence by `rowSums` in `.data_core`, build the structural-zero table directly in `.get_struc_zero`, draw the t variates of `.dunn_global` for the whole matrix of degrees of freedom in one call from the same stream, preallocate the output of `.combn_fun2`, and avoid repeated transposition of the feature table in `.ancombc2_core`.

ancombc2, random effects

`.lmer_struct_all` groups taxa by their pattern of non-missing responses and calls `lme4::lFormula` once per group. The resulting structure, which holds the model frame, the fixed-effects model matrix and the random-effects terms, does not depend on the response values and is reused by every taxon and every REML iteration in that group.

`.lmer_refit` fits one response from a stored structure by the steps that `lme4::lmer` performs after `lFormula`: `mkLmerDevfun`, `optimizeLmer`, `checkConv` and `mkMerMod`. The covariance parameters `reTrms$theta` and the nonzero entries of `reTrms$Lambda` are restored to their starting values in a private copy before each fit, because `lme4` modifies them in place during optimization. Each fit therefore does not depend on the fits that precede it.

`lmerTest::lmer` is called only where Satterthwaite degrees of freedom are required, in the final fit at the converged covariance parameters, and `summary()` is called once per taxon. A taxon whose variance-covariance matrix is not positive definite is reported as unfitted. The initial pass and the first REML iteration both evaluate the model at $\theta = 0$, and one pass serves both. From each fit, the coefficients, fitted values, residuals and variance-covariance matrix are extracted and the `merMod` object is discarded.

secom_linear

The bodies of the `foreach` loops are evaluated in an environment constructed by `.detached_env`, whose parent is the global environment and whose contents are the matrices and scalars the body reads, together with `foreach` and `%dorong%`. The parallel backend therefore loads no package namespace on the workers at each call. The `%dorong%` streams are unchanged.

The co-occurrence matrix is the cross-product of the occurrence indicator matrix, with its diagonal set to the column sums of that indicator.

`.thresh_loss_grid` evaluates the cross-validation loss at every point of the threshold grid in one pass. The entries of the training correlation matrix are sorted once by absolute value; `findInterval` gives the number of entries set to zero at each threshold; and the loss follows from prefix sums of b^2 and suffix sums of $(a - b)^2$, with the additional suffix sums that soft thresholding and the weighted penalty require.

The false-positive screen computes the correlation of the estimated sampling-fraction difference with the columns of the abundance matrix and forms the mask by `outer` on the resulting indicator.

secom_dist

`secom_dist` uses the same detached evaluation environment for its `foreach` loop, the same cross-product form of the co-occurrence matrix and the same false-positive screen as `secom_linear`. The distance correlation and its p-value are both taken from the single `energy::dcor.test` call for a pair of taxa.

5 The conservative argument

The sensitivity analysis to pseudo-count addition compares the significance of a taxon on the complete data with its significance under added pseudo-counts. The argument `conservative`

selects the rule, and applies only when `pseudo_sens = TRUE`. Let α denote the significance level and $\mathbb{I}\{\cdot\}$ the indicator function. The rules below are applied column by column to `res`, `res_global`, `res_pair`, `res_dunn` and `res_trend`, which carry the same column layout under both settings.

`conservative = TRUE (default)`. The pseudo-count set is

$$\mathcal{P}_1 = \{0.1, 0.5, 1\}. \quad (2)$$

Each value is added to the feature table before the estimation of the sampling fractions, and the ANCOM-BC2 algorithm is re-run in full. Let $q^{(c)}$ denote the adjusted p-value of a taxon in the run with pseudo-count c , with $q^{(0)}$ the adjusted p-value of the run on the complete data. Then

$$\text{score} = \frac{1}{|\mathcal{P}_1| + 1} \sum_{c \in \{0\} \cup \mathcal{P}_1} \mathbb{I}\{q^{(c)} > \alpha\}, \quad (3)$$

$$\text{passed_ss} = \mathbb{I}\{\text{score} \in \{0, 1\}\}, \quad (4)$$

$$\text{diff_robust} = \text{diff} \wedge \text{passed_ss}. \quad (5)$$

`conservative = FALSE`. The sampling fractions are estimated once, on the complete data. The pseudo-count set is

$$\mathcal{P}_0 = \{0.01, 0.02, \dots, 0.50\}, \quad |\mathcal{P}_0| = 50. \quad (6)$$

Each value is added to the zero entries of the bias-corrected table, and the fixed effects are re-estimated by ordinary least squares, one fit per taxon per pseudo-count, distributed over the `%dorng%` loop. Let $p^{(c)}$ denote the unadjusted p-value of a taxon at pseudo-count c , and $p^{(0)}$ the unadjusted p-value of the run on the complete data. Then

$$\text{score} = \frac{1}{|\mathcal{P}_0|} \sum_{c \in \mathcal{P}_0} \mathbb{I}\{p^{(c)} > \alpha\}, \quad (7)$$

$$\text{passed_ss} = \mathbb{I}\{\text{score} = 0, p^{(0)} \leq \alpha\} + \mathbb{I}\{\text{score} = 1, p^{(0)} > \alpha\}, \quad (8)$$

$$\text{diff_robust} = \text{diff} \wedge \text{passed_ss}. \quad (9)$$

A taxon with $p^{(0)}$ missing is assigned $p^{(0)} = 1$. This rule is the one used in Bioconductor release 3.19.

Example counts. On the QMP simulation of the ANCOMBC2 vignette (91 taxa), the three `res_pair` comparisons give 7, 7 and 12 taxa with `diff_robust = TRUE` under both settings. The counts of `passed_ss = TRUE` are 72, 70 and 90 with `conservative = FALSE` and 67, 55 and 84 with `conservative = TRUE`. On `dietswap` at the Family level, the `nationalityAFR` column of `res` gives 4 taxa with `diff_robust = TRUE` under `conservative = TRUE` and 5 under `conservative = FALSE`; `res_global` gives 3 and 2, and `res_trend` gives 5 and 3. The direction of the `res_trend` difference follows from the test used in the refits: the sequential F test of the ordinary least squares refit is not the constrained trend test of the main run, and for a repeated-measures design the two disagree on a subset of taxa. On `dietswap`, the sensitivity analysis took 239 s with `conservative = TRUE` and 77 s with `conservative = FALSE`, measured on the installation preceding the random-effects changes of Section 2.

6 Bug fixes

`ancombc with pseudo = 0`. The response of the fixed-effects fit is

$$y_{ij} = \log(o_{ij} + \text{pseudo}), \quad y_{ij} \leftarrow \text{NA} \text{ if } y_{ij} \text{ is not finite}, \quad (10)$$

where o_{ij} is the count of taxon i in sample j . With `pseudo = 0` and $o_{ij} = 0$ the first expression gives $-\infty$, and the second assignment treats that entry as missing, as `.ancombc2_core` does. The taxon is then fitted on its observed samples. In case 1 of `benchmarks/harness/edge_cases.R`, a table of 60 taxa and 60 samples with zero counts in two taxa, three taxa that 2.13.2 returns as NA are estimated, and no entry of `diff_abn` changes.

`ancombc2` with `trend = TRUE` and `global = FALSE`. The sensitivity analysis of the trend test reads the sensitivity table of the global test. When `trend = TRUE`, the global test is therefore computed regardless of the value of `global`, and its result is removed from the return value when `global = FALSE`. The combination `pseudo_sens = TRUE`, `trend = TRUE` and `global = FALSE`, which stopped with 'MARGIN' does not match `dim(X)`, now returns the trend results with their `passed_ss` and `diff_robust_abn` columns and no `res_global` component.

7 Remaining costs

The shares below are from Rprof profiles of version 2.15.1.

- In `secom_dist_atlas`, the compiled permutation loop of `energy::dcor.test`, entered through `.C`, accounts for 95.1% of self time. The cost is $O(d^2 R)$ permutation replicates for d taxa.
- In `ancombc2_dietswap`, the random-effects path `.remle_fit_all` accounts for 86.2% of total time, and the `lme4` optimizer called through `nloptr` accounts for 39.0% of total time, of which the objective function `eval_f` accounts for 30.4%.
- In `ancombc_atlas_genus_large`, the `nloptr` call of the M step in `.bias_em` accounts for 72.9% of total time; its objective function accounts for 54.2% of total time and 22.9% of self time, and the `dnorm` evaluations of the E step account for a further 22.9% of self time.
- In the fixed-effects profiles, `.sandwich_vcov` accounts for 4% to 19% of self time.

8 Reproducing the measurements

The scripts take all paths from their command-line arguments. Let `lib_a` and `lib_b` denote library directories holding the two installations to compare, and `out_a` and `out_b` the directories for the results. From the package root:

```
Rscript benchmarks/harness/run.R      lib_a out_a all 3
Rscript benchmarks/harness/run.R      lib_b out_b all 3
Rscript benchmarks/harness/compare.R  out_a out_b 1e-8
Rscript benchmarks/harness/compare.R  out_a out_b 0
```

`run.R` takes the library path, the output directory, the scale (`all`, `vignette` or `large`), the number of repetitions, and an optional regular expression restricting the workload names. It writes one `.rds` file per workload and `timings.csv`. `compare.R` takes the two output directories and the tolerance, and writes `compare_vs_<basename(out_a)>.csv` into the second directory. The edge cases and the `pseudo = 0` reproduction are run as

```
Rscript benchmarks/harness/edge_cases.R  lib_b out_b/edge_cases.rds
Rscript benchmarks/harness/repro_pseudo0.R lib_b
```